Ameya Deshmukh

Boston, MA(Open to Relocate) | +1 (857)3967408| deshmukh.amey@northeastern.edu | LinkedIn | GitHub | Portfolio

EDUCATION

Northeastern University, Boston, MA — Expected Dec 2025

Master of Science in Data Analytics Engineering — **GPA: 3.7/4.00**

Coursework: Statistical methods (A/B testing), Data Management, Advanced NLP, Data mining

Vishwakarma Institute of Information Technology, Pune, India — Aug 2018 - May 2022

Bachelor of Engineering in Electronics and Telecommunication — **GPA: 3.76/4.00**

Coursework: Machine Learning, Deep learning, Data Structures and Algorithms

Technical Skills

Programming Languages:	C++, Python (Pandas, NumPy, Scikit-learn), R, **SQL**, NoSQL, Linux
Databases:	MySQL, PostgreSQL, RDBMS, Snowflake
Data Visualization Tools:	Tableau, Power BI, Microsoft Excel, Looker, Qlik, Alteryx, MATLAB
Data Engineering:	Spark, Hadoop, Git, CI/CD, DevOps, DataBricks
Data Science:	Deep Learning, TensorFlow, Pytorch, Machine Learning, Scikit-learn, SciPy
Cloud and Infrastructure:	AWS, Redshift, S3, EC2, Lambda, Google Cloud, Azure

Work Experience

Analytics and Insights Gen AI Co-op, Boehringer Ingelheim, CT — **Jan 2025 – July 2025**

Python| SQL| ThoughtSpot| OpenAI-API| K-means clustering| SHA-encryption

- Engineered RAG-based Q&A bot for unstructured healthcare data using k-means clustering to index 768-dimensional ada-002 embeddings, reducing query latency from **5s to 2s** while maintaining **85%** answer relevance.
- Standardized and structured payer-plan landscape information by implementing **fuzzy-matching** using sequence-matcher libraries improving **data consistency and downstream analytics**.
- Collaborated with two cross-functional teams to conduct **exploratory data analysis** using **Gen-AI**, identifying statistical trends in KPIs such as regional performance metrics and 3-month lookback performance.

Office Data Engineer, Northeastern University, MA — **May 2024 – Nov 2024**

Python| Excel| GCP| Flask| Github| Visual studio| Agile| HTML

- Optimized **ELT pipelines** with **Airflow** scheduling for curated website articles, **reducing data update time by 25%**.
- Maintained and enhanced functionality for websites built on **Flask** and **React** improving UI (CSS) and stability.
- Managed **GCP** Datastores, Cloud Storage, and **version control** ensuring streamlined data management and website uptime.
- Utilized **Trello and GitHub** with cross functional agile team for data driven troubleshooting and code management**.**

Data Analyst Intern, Newtuple, India — **Jan 2023 – May 2023**

Python| LLM's| SQL| AWS EC2| REST API| Langchain| OpenAI| ***DBT****| Looker*

- Developed an AI chat-bot for an e-commerce platform using **Python and DBT** reducing business analysis time by 30%.
- Automated **ETL** data pipelines for **ad-hoc data analysis** on dashboards, achieving **30% time** savings for pipelining.
- Utilized prompt engineering using **LLMs (Langchain)** converting questions to **SQL** queries, simplifying database access.
- Cleaned and optimized data models ('inventory', 'customers', 'orders') in **PostgreSQL** for **AWS EC2** deployment.

Projects

Transformer Language Model (Warren Buffett's letters dataset**)** | Pytorch, Softmax, cross-entropy, NLP | **Link**

- Architected a 256-dimensional, 4-layer transformer encoder with causal masking in PyTorch with 5.5x loss reduction.
- Built end-to-end MLOps pipeline using Docker, ECR, and S3 for reproducible model training on AWS SageMaker.
- Implemented multi-head attention mechanism (8 heads) with positional embeddings achieving 0.21 validation loss.

Online News Popularity Classification |*Machine Learning, PCA, SVM, Python* **| Link**

- Built a **custom SVM** classifier in Python using Sequential Minimal Optimization algorithm.
- Performed data validation & addressed highly correlated columns to improve data quality by performing data analysis.
- Applied cubed root transformation to handle right-skewed distributions performing data manipulation using ANNOVA.
- Reduced dataset dimensions from **27 to 15** features using **PCA**, capturing essential variance.

Finance – Market Analysis |*Trend analysis, Predictive modelling, Prophet,LSTM* **| Link**

- Applied stock price path finding techniques using mathematical models using python (**Prophet**) scripting.
- Used Monte Carlo Simulation optimization to generate basic sampling for forecasting models using statistical techniques.
- Conducted financial forecasting for time-series using models like '**ARIMA**' and deep learning models like '**LSTM**'.

Travel Data Management Application |MySQL, NoSQL | **Link**

- Performed Advanced analytics using SQL Queries like CTEs and subqueries for optimized retrieval of desired results.
- Built a customer-facing web app using 'Streamlit' in python connected to MySQL database capturing customer data.
- Collected structured and unstructured data using NoSQL for visualization and finding solutions.